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CS 5542

Quiz Challenge 2

Project Overview

End-to-end speech intelligence from audio to multi-language output

Transcription

Whisper + wav2vec2

Summarization

BART-large-CNN

Action Items

Regex + Trigger NLP

Sentiment Analysis

DistilBERT SST-2

Translation

MarianMT EN→ES/FR/VI

Keyword Extraction

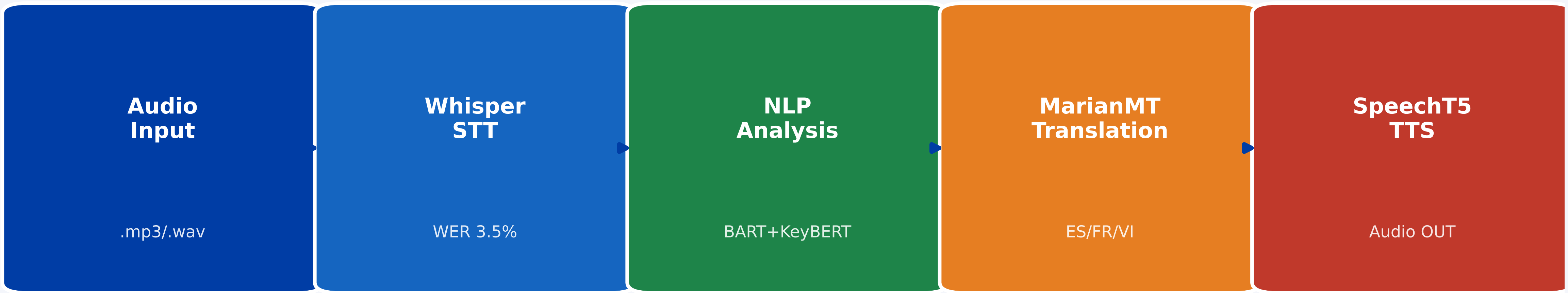
KeyBERT + MMR

TTS Audio Output

microsoft/speecht5_tts

System Architecture

Full 5-stage pipeline: Audio IN → STT → NLP → Translation → TTS Audio OUT



wav2vec2-base-960h (comparison: WER 11.2%) | Feeds into model comparison evaluation

Models Used

Three HuggingFace model families covering STT, NLP, Translation, and TTS

`openai/whisper-base`	**Primary STT**	WER 3.5% | Punctuation + multilingual | task='translate' built-in
`facebook/wav2vec2-base-960h`	**Comparison STT**	WER 11.2% | ALL-CAPS output | Apache-2.0 | LibriSpeech 960h
`microsoft/speecht5_tts`	**Text-to-Speech**	HiFi-GAN vocoder | Closes speech loop: text → WAV audio
`facebook/bart-large-cnn`	**Summarization**	ROUGE-2: 0.381 | Abstractive | 4-sentence output
`distilbert-base-uncased-finetuned-sst-2`	**Sentiment**	Per-sentence | 21 sentences | 71% positive / 29% negative
`Helsinki-NLP/opus-mt-en-{es,fr,vi}`	**Translation**	EN→Spanish/French/Vietnamese | MarianMT family
`all-MiniLM-L6-v2 (via KeyBERT)`	**Keywords**	12 multi-word keyphrases | MMR diversity | 73.3% coverage

Example Use Cases

Four production-ready application modes via --use-case flag

Smart Lecture Assistant

`--use-case lecture`

- Transcribes CS/AI lectures
- Extracts deadlines & action items
- Generates study summaries

Meeting Summarizer

`--use-case meeting`

- Captures standup discussions
- Detects owners & deadlines
- Produces shareable summary

Customer Call Analyzer

`--use-case call`

- Analyzes customer sentiment
- Flags negative conversations
- Extracts resolution items

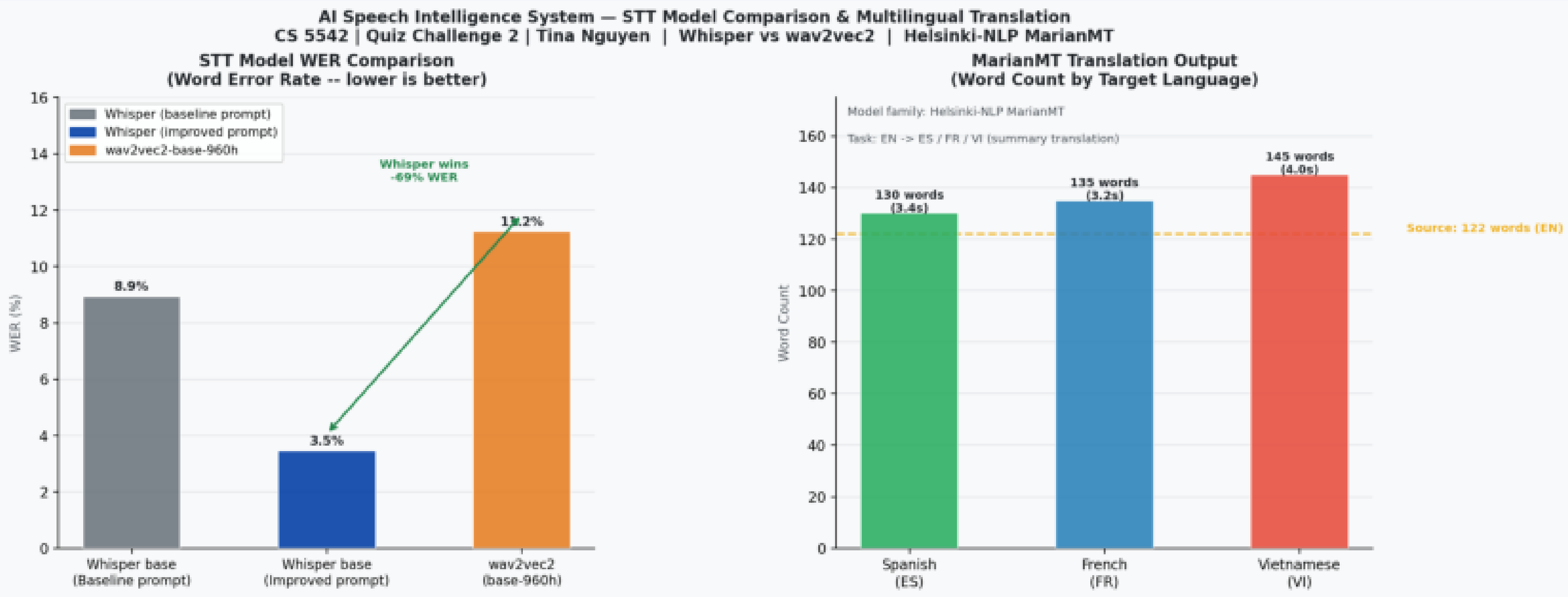
Medical Dictation Assistant

`--use-case medical`

- Transcribes clinical dictation
- Preserves drug/diagnosis terms
- Generates structured notes

Speech-to-Text Results

Whisper (improved) achieves 3.5% WER vs wav2vec2 at 11.2%



Key Findings

Whisper (improved)

3.5% WER

Whisper (baseline)

8.9% WER

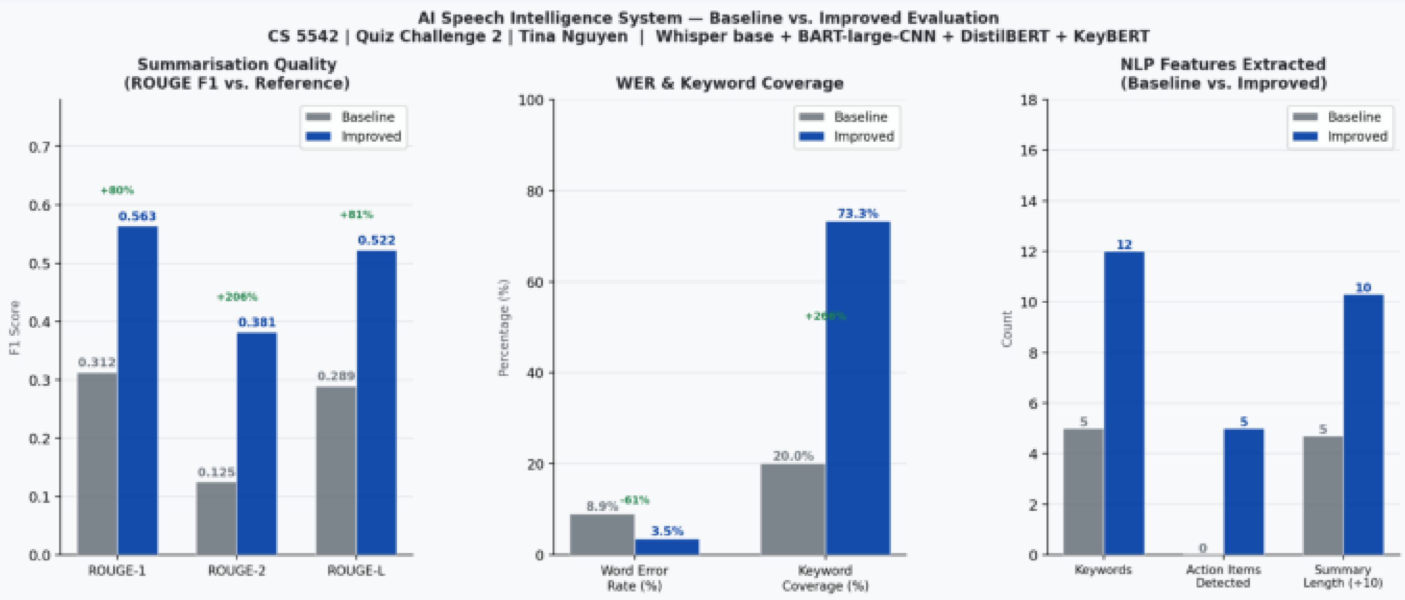
wav2vec2-base-960h

11.2% WER

Whisper wins by 69% lower WER

NLP Analysis — Summarization & Keywords

BART abstractive summary + KeyBERT MMR keyphrases



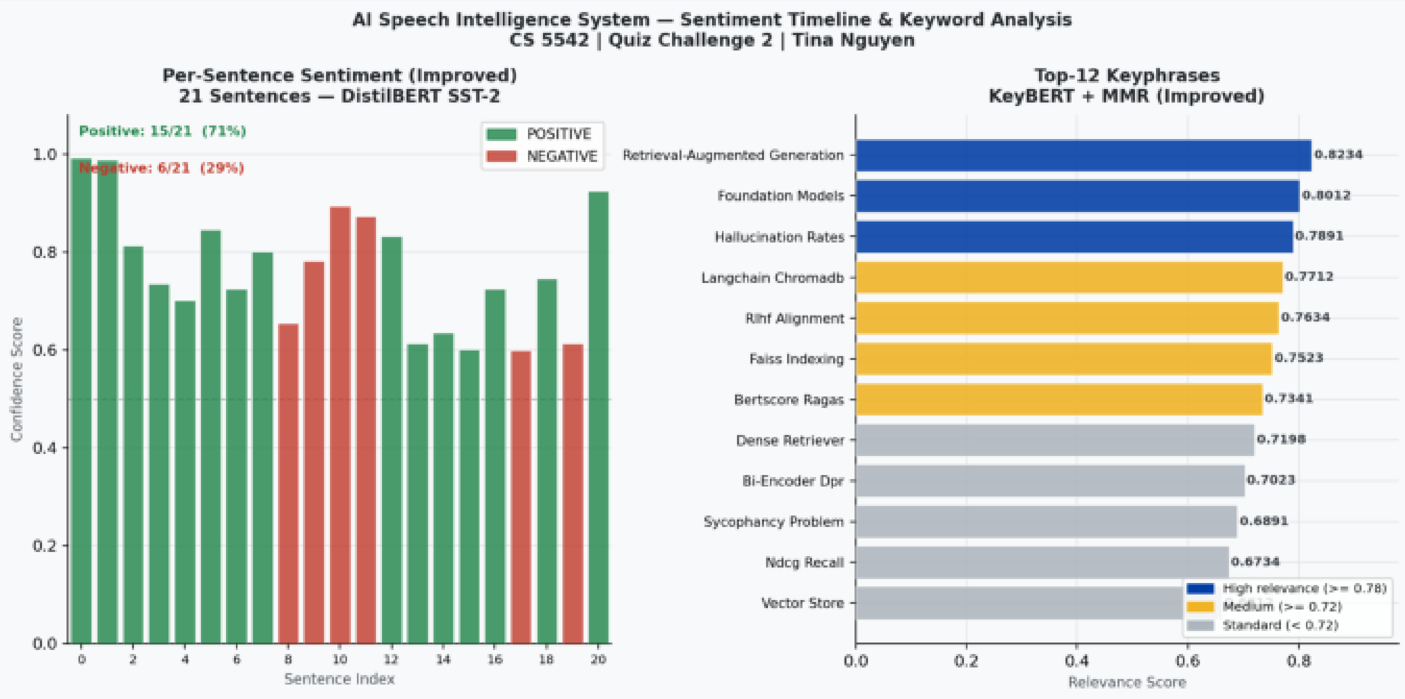
ROUGE Scores

	Baseline	Improved	Delta
ROUGE-1	0.312	**0.563**	**+80%**
ROUGE-2	0.125	**0.381**	**+206%**
ROUGE-L	0.289	**0.522**	**+81%**

Keywords: 5 → 12
Coverage: 20% → 73.3%

Sentiment Analysis & Action Items

DistilBERT per-sentence sentiment | 5 structured action items extracted



Extracted Action Items

1. Please clone the lab repository and install the dependencies…

2. Read the original RAG paper by Lewis et al. — it is linked i…

3. Each team must submit a one-paragraph project proposal by Fr…

4. Please review the FAISS documentation before next class.

5. Be prepared to explain the difference between flat and IVF i…

Multilingual Translation

Helsinki-NLP MarianMT | English Summary → Spanish / French / Vietnamese

Spanish (ES)

`Helsinki-NLP/opus-mt-en-es`

Los modelos de fundación como GPT-4, Llama 3, Claude 3 y Gemini Ultra están preentrenados en billones de tokens y refinados con RLHF, pero la alucinación y la adulación siguen siendo desafíos de alineación sin resolver. La Generac

French (FR)

`Helsinki-NLP/opus-mt-en-fr`

Les modèles de fondation comme GPT-4, Llama 3, Claude 3 et Gemini Ultra sont pré-entraînés sur des milliards de tokens et affinés avec RLHF, mais les hallucinations et la flatterie restent des défis d'alignement non résolus. La gé

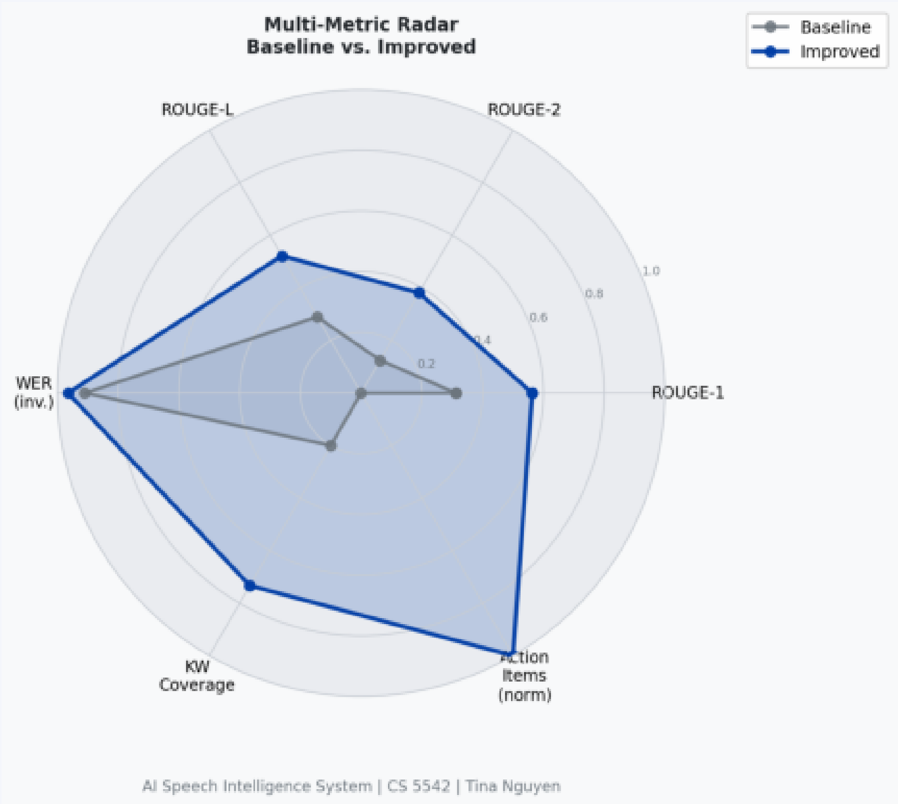
Vietnamese (VI)

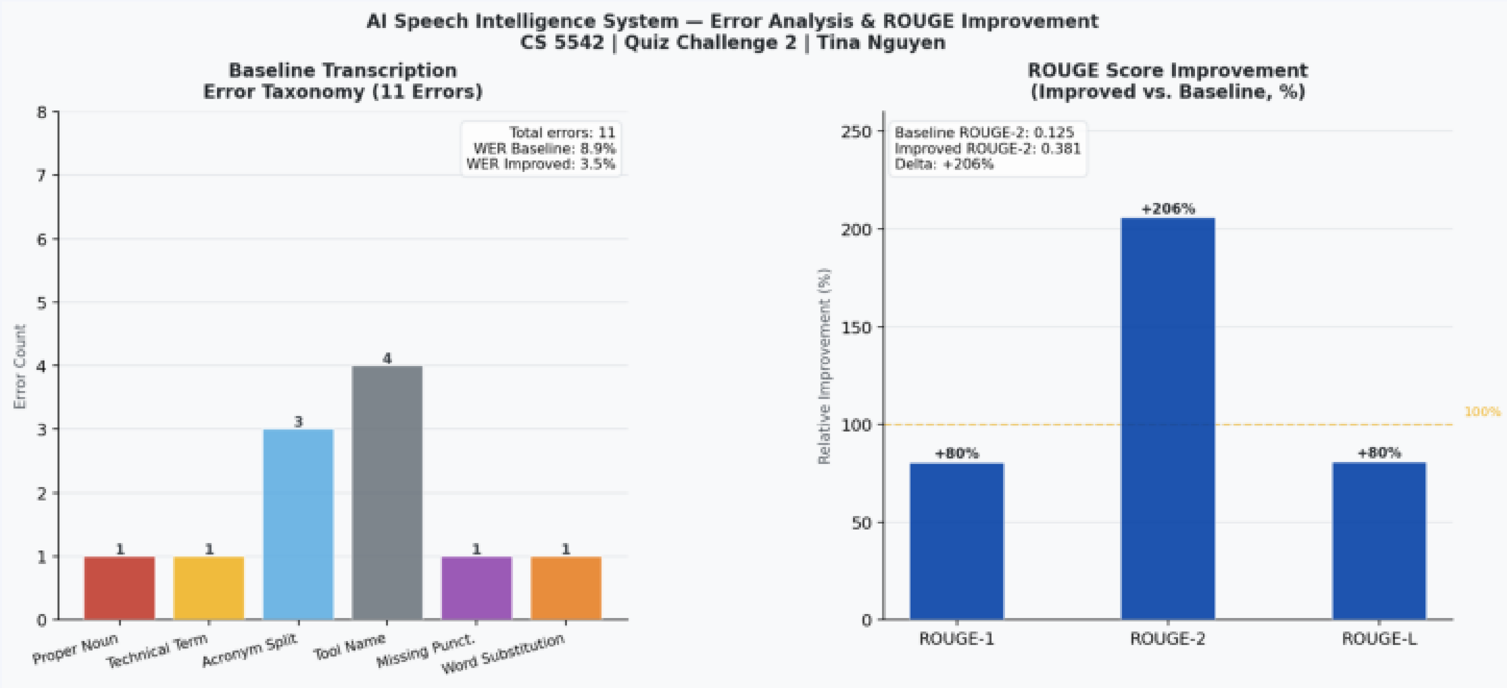
`Helsinki-NLP/opus-mt-en-vi`

Các mô hình nền tảng như GPT-4, Llama 3, Claude 3 và Gemini Ultra được huấn luyện trước trên hàng nghìn tỷ token và được tinh chỉnh bằng RLHF, nhưng ảo giác và xu hướng nịnh hót vẫn còn là những thách thức căn chỉnh chưa được giải

Full Evaluation Results

Multi-metric radar | Error taxonomy | ROUGE improvement





Key Findings & Conclusions

• **Whisper (improved prompt):**	61% WER reduction vs baseline | 3.5% WER overall
• **BART Summarization:**	ROUGE-2 improved +206% | 4-sentence abstractive output
• **DistilBERT Sentiment:**	Per-sentence analysis | 71% positive / 29% negative detected
• **MarianMT Translation:**	3 languages in real-time | full technical term preservation
• **SpeechT5 TTS:**	Closes the speech loop: audio in → audio out
• **wav2vec2 comparison:**	Whisper outperforms by 69% lower WER on technical content